Transfer Learning Improves Landslide Susceptibility Mapping in Data-scarce Landscapes.

Rachel Ayuma

Abstract

Landslides pose significant threats to communities, particularly in mountainous regions of developing countries where data for hazard assessment is often scarce. This study explores the application of cross-regional transfer learning (TL) for landslide susceptibility modeling (LSM), leveraging a data-rich region ,West Pokot, as the source, and a data-scarce region, Elgeyo Marakwet, as the target. Using nine time-independent landslide conditioning factors derived from remote sensing and GIS data, a Support Vector Machine (SVM) model was trained in West Pokot and directly transferred to Elgeyo Marakwet. The model's predictive performance was evaluated using accuracy, F1-score, kappa coefficient, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). In West Pokot, the model achieved high accuracy (91.53%) and F1-score (91.06%), while in Elgeyo Marakwet, performance was slightly lower (83.33% accuracy and 80.0% F1-score) due to regional differences in terrain and hydrology. Results confirm the viability of transfer learning for enhancing landslide prediction in data-deficient regions, reducing the reliance on extensive local inventories and improving disaster preparedness through cost-effective spatial analysis.

Keywords: Landslides, Transfer Learning, Support Vector Machine, Landslide Susceptibility Mapping, Remote Sensing, GIS, Machine Learning

1. Introduction

Landslides are a major natural hazard commonly associated with mountainous regions, often triggered by earthquakes and intense rainfall. These events primarily affect mountainous areas worldwide, significantly impacting local communities and economies (Shrestha et al., 2025). According to data from the World Health Organization (WHO), between 1997 and 2017, approximately 4.7 million people were affected, and around 56,000 individuals lost their lives due to landslides. The frequency and severity of landslides tend to be higher in developing nations, largely due to inadequate susceptibility mapping and insufficient risk assessment (Sim et al., 2022).

The development of landslide susceptibility modeling (LSM) has progressed significantly since the early 1970s. Over time, the methodology has evolved from deterministic models to heuristic, statistical, and, more recently, machine learning approaches (Wang et al., 2022). Initially, deterministic methods were widely used but were often subjective and limited in predictive capability. With the advancement of remote sensing technology, heuristic and statistical models gained popularity. However, these approaches struggled to capture the complexity and non-linear nature of landslide occurrences. Consequently, modern machine learning techniques including Logistic Regression (LR), Support Vector Machines (SVM), Random Forests

(RF), Multi-layer Perceptrons (MLP), and Deep Neural Networks (DNN) are now extensively applied for landslide prediction (Singh et al.,2024). These models function by utilizing landslide conditioning factors as independent variables and landslide inventories as dependent variables to generate probability maps of landslide-prone locations.

This study focuses on Transfer learning (TL) a technique that allows a model trained on one dataset to be adapted for use in another, minimizing the need for extensive new data collection. TL has already been successfully applied in various fields, such as land use classification, water resource management, vegetation analysis, crop yield prediction, and natural hazard assessment (Singh et al.,2024). In landslide prediction, where data scarcity is a major challenge, TL offers a promising alternative.

To address the existing limitations of landslide susceptibility mapping (LSM) in data-scarce regions, this study proposes a transfer learning (TL) approach combined with a Support Vector Machine (SVM) model to identify landslide-prone areas. The key research gap lies in the limited availability of high-quality landslide inventory data in regions like Elgeyo Marakwet, which hampers the development of accurate and generalizable machine learning models. To overcome this, the study introduces a novel cross-regional TL framework that transfers knowledge from West Pokot County, a data-rich region with a well-documented history of landslides, to Elgeyo Marakwet, where landslide data is scarce. The main objective is to enhance LSM accuracy in the target region without the need for extensive new data collection. The methodology involves training the SVM model on the source region and applying it to the target region to generate reliable LSMs. The performance is evaluated using statistical metrics such as precision, recall, F-score, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Practically, the findings aim to support disaster risk reduction, inform land-use planning, and provide decision-makers with a cost-effective and scalable tool for improving landslide preparedness in data-limited environments.

2. Materials and Methods

2.1 Study Areas

2.1.1 West Pokot County (Source Region)

West Pokot County is located in northwestern Kenya, covering approximately 9,169.4 km² (Figure 1). Geographically, it lies between latitudes 1° and 2° North and longitudes 34° 47′ and 35° 49′ East. The region is characterized by a diverse physiographic landscape, ranging from low-lying semi-arid zones to highland areas with steep slopes and rugged terrain. The geology of the area includes Precambrian basement rocks such as gneisses and schists, along with volcanic formations, making the area geologically unstable. The area is frequently affected by landslides, especially during periods of intense rainfall. Historical records and inventories from West Pokot provide a rich dataset for machine learning model development.The socio-economic context is largely rural, with agriculture and livestock rearing being the primary economic activities. The presence of the Cherangani Hills, a major water catchment area, exacerbates the landslide risk due to deforestation, soil erosion, and water saturation during rainy seasons.

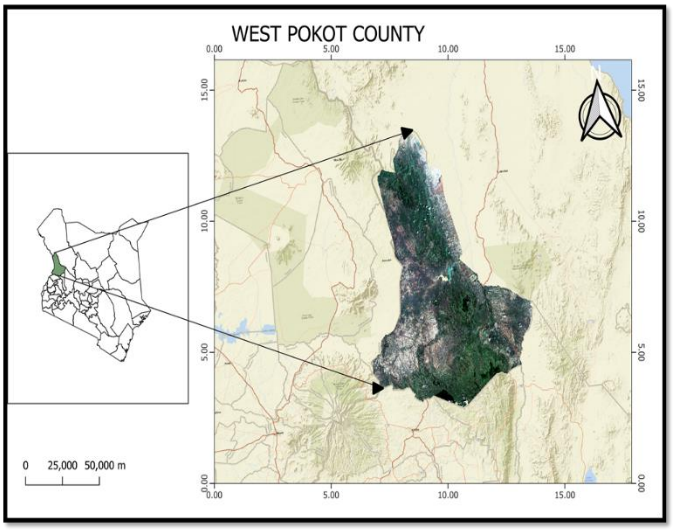


Figure 1: Map of the source region (West Pokot County)

2.1.2 Elgeyo Marakwet County (Target Region)

Elgeyo Marakwet County lies within the Great Rift Valley in western Kenya (Figure 2). Geographically, it is situated between latitudes 0° 10′ N and 1° 30′ N and longitudes 35° 25′ E and 35° 45′ E. It is bounded by diverse topographical and ecological features, including the Kerio Valley and the Cherangani Hills. Elevation varies from 900 m in the valley to over 3,350 m in the highlands. Geologically, the region comprises volcanic deposits, metamorphic rocks, and Quaternary sediments, contributing to high slope instability.The area is known for frequent landslide events, particularly in regions with steep escarpments and heavy rainfall. Despite these hazards, comprehensive landslide inventories are lacking, limiting the effectiveness of traditional LSM techniques. This makes Elgeyo Marakwet an ideal candidate for transfer learning applications

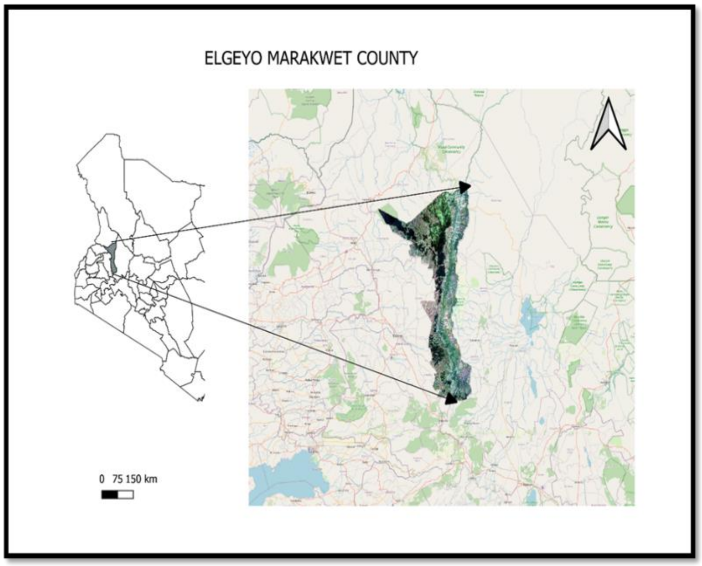


Figure 2: Map of the target region (Elgeyo Marakwet County)

2.2 Data Sources

This study utilized a combination of remote sensing and GIS datasets to support landslide susceptibility modeling. Topographic data was obtained from the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) at 30-meter resolution. Key terrain derivatives, elevation, slope, aspect, curvature, and the Topographic Wetness Index (TWI), were computed from the DEM to assess slope stability.

Vegetation cover was represented using the Normalized Difference Vegetation Index (NDVI), derived from Landsat 8 Operational Land Imager (OLI) imagery, processed via Google Earth Engine (GEE). NDVI served as a proxy for vegetation density, which plays a role in soil cohesion and hydrological regulation.

Landslide inventory data for West Pokot, the source region, were obtained from the Global Precipitation Measurement (GPM) Mission, comprising 429 landslide points. Elgeyo Marakwet, the target region, had only 11 recorded landslide events. Lithological data were sourced from the Water Resource Authority (WRA), while vector data for rivers and roads were extracted from OpenStreetMap and DIVA-GIS to evaluate the impact of hydrology and infrastructure on landslide occurrence.

All spatial layers were standardized to 30-meter resolution and aligned to a common coordinate system. Data preprocessing and analysis were conducted using QGIS, ArcGIS, R, and GEE, ensuring consistency and replicability.

2.3 Methodology

2.3.1 Theory of Transfer Learning

The performance of machine learning models is often constrained by a lack of labeled datasets, limiting their predictive capabilities. Transfer Learning (TL) mitigates this challenge by leveraging knowledge from a pre-trained model on a related task (source domain) and applying it to a new task (target domain) (Wang et al., 2023). This is particularly beneficial when the target domain has insufficient data for training a model from scratch.

TL operates between two domains: the source domain, where data is abundant and well-structured, and the target domain, where data is scarce (Wang et al., 2023). Based on how knowledge is transferred, TL is categorized into four types: instance transfer, which assigns weights to source data instances; relational-knowledge transfer, which captures shared patterns between domains; feature representation transfer, which aligns feature spaces for better generalization; and parameter transfer, where learned parameters from a well-trained model are reused to improve learning in the target domain.

Unlike traditional machine learning, which requires extensive labeled data, TL reduces data dependency and computational costs by reusing knowledge from related tasks. This is particularly useful in landslide susceptibility prediction, where data availability varies across regions (Singh et al., 2024). Fine-tuning a pre-trained model with TL is faster and more efficient than training from scratch, enabling strong performance even with limited data.

By transferring insights from data-rich areas to regions with limited historical records, TL enhances predictive accuracy and accelerates model training. This makes it a valuable tool for addressing data scarcity challenges in machine learning applications, especially in geographically diverse tasks like landslide prediction.

2.3.2 Preparation of Landslide and Non-landslide Datasets

A landslide inventory provides the foundational data required for training susceptibility models, offering spatial information on past landslide events. These inventories are often developed through a combination of field surveys, remote sensing analysis, and visual interpretation of satellite imagery (Singh et al., 2024). Point-based landslide inventories have become increasingly preferred over polygon-based formats due to their enhanced performance in machine learning models. In this study, point-based landslide data were sourced from the Global Precipitation Measurement (GPM) Mission, resulting in 429 landslide events in West Pokot and 11 in Elgeyo Marakwet counties.

To train a supervised classification model, both landslide (positive) and non-landslide (negative) samples are required. While positive samples are obtained directly from the inventory, the selection of non-landslide points must be done carefully to avoid sampling from areas that are potentially unstable. Several sampling methods exist for this purpose, including Mahalanobis distance, frequency ratio, and buffer-based exclusion (Guo et al., 2024). However, this study employed random sampling with appropriate spatial buffers around known landslides to minimize the risk of mislabeling unstable areas.

Random sampling was chosen due to its simplicity, computational efficiency, and wide use in regional-scale susceptibility modeling (Guo et al., 2024). Though it does not account for all spatial complexities, the method was implemented with safeguards to ensure that non-landslide points were only drawn from stable regions with no recorded landslide activity. This approach provided a balanced and practical solution for generating negative samples. A standard 70:30 train-test split was applied to the final dataset to facilitate model training and validation.

2.3.3 Landslide Conditioning Factors

Landslide Conditioning Factors (LCFs) are typically classified into time-dependent and time-independent variables. Time-independent factors remain relatively constant over time and are more suitable for regional-scale landslide susceptibility mapping. In contrast, time-dependent factors, such as rainfall and land use, vary across seasons and regions, potentially introducing inconsistencies in model performance (Singh et al., 2024).

This study focuses exclusively on time-independent LCFs to ensure consistency across the source and target regions, which differ significantly in terms of rainfall and land cover dynamics (Zhao et al., 2024). Nine key factors were selected based on their relevance to slope stability and availability of data: slope, aspect, curvature, elevation, distance to rivers, distance to roads, lithology, topographic wetness index (TWI), and normalized difference vegetation index (NDVI). These factors were derived from a combination of remote sensing datasets and processed using ArcGIS, QGIS and Google Earth Engine.

2.3.3.1 Slope

Slope angle is a critical determinant of gravitational force acting on slope materials. Steeper slopes are generally more prone to landslides. Slope (in degrees) was computed from the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) using QGIS. The 30-meter resolution DEM was processed using the terrain analysis function in QGIS to generate slope maps. These values were categorized into susceptibility classes using natural breaks (Jenks) for analysis.

2.3.3.2 Aspect

Aspect refers to the compass direction a slope faces and influences factors like sunlight exposure, vegetation cover, and evapotranspiration. Aspect was derived from the DEM using QGIS's raster terrain analysis tool and classified into 8 cardinal directions (N, NE, E, SE, S, SW, W, NW) along with a flat category.

2.3.3.3 Curvature

Curvature represents the convexity or concavity of the terrain, influencing water flow and sediment transport. Profile and plan curvature were computed from the DEM using QGIS, and then merged into a general curvature layer. Concave surfaces tend to collect water, increasing susceptibility, while convex surfaces are generally more stable.

2.3.3.4 Elevation

Elevation affects climatic conditions, vegetation, and geological weathering. It was directly extracted from the SRTM DEM and maintained in continuous format for the model. Higher altitudes tend to experience different erosion and precipitation regimes that affect slope stability.

2.3.3.5 Distance to Roads

Road construction often destabilizes slopes through excavation and drainage disruption. Distance to roads was calculated using Euclidean distance from digitized road layers sourced from OpenStreetMap (OSM) and DIVA-GIS. The resulting raster values (in meters) represent proximity to roads and were used as continuous inputs in the model.

2.3.3.6 Distance to Rivers

Rivers cause toe erosion and increase soil saturation, contributing to landslide risk. Distance to rivers was derived by computing Euclidean distance from hydrology shapefiles (streams and rivers) obtained from DIVA-GIS. The raster values (in meters) were used to reflect proximity to fluvial activity.

2.3.3.7 Normalized Difference Vegetation Index (NDVI)

NDVI indicates vegetation density and is closely linked to slope stability through root reinforcement and moisture regulation. NDVI was derived from Landsat 8 OLI imagery using the formula:

$$NDVI = \frac{NIR-RED}{(NIR+RED)} \text{.....................(1)}$$

Where NIR is Band 5 and RED is Band 4. Google Earth Engine (GEE) was used for image preprocessing and NDVI calculation. Higher NDVI values generally indicate more stable, vegetated slopes.

2.3.3.8 Topographic Wetness Index (TWI)

TWI quantifies the tendency of a location to accumulate water and is expressed as:

$$TWI = \ln\left(\frac{a}{\tan\beta}\right) \text{.....................(2)}$$

Where:

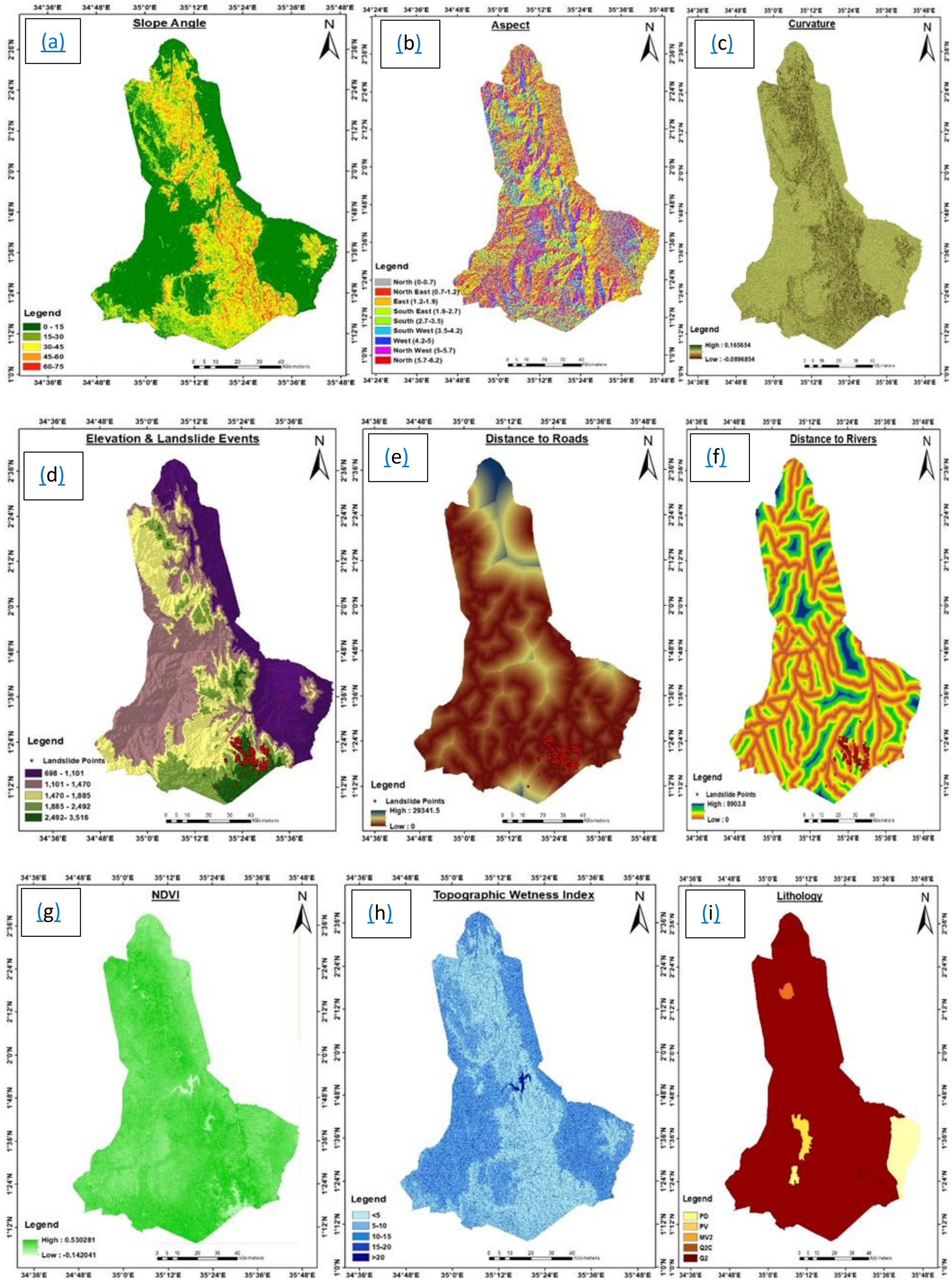
- Specific Catchment Area (a): This represents the area of land contributing water flow to a specific point, normalized by the flow width.
- Slope Angle (β): This is the angle of the slope at a particular location, measured in radians.
- ln: This represents the natural logarithm, used to scale the TWI values.

2.3.3.9 Lithology

Lithology describes the geological composition of an area, which affects permeability, shear strength, and weathering. Lithological maps were sourced from the Water Resources Authority and digitized into thematic layers representing rock types. These were encoded as categorical variables for modeling.

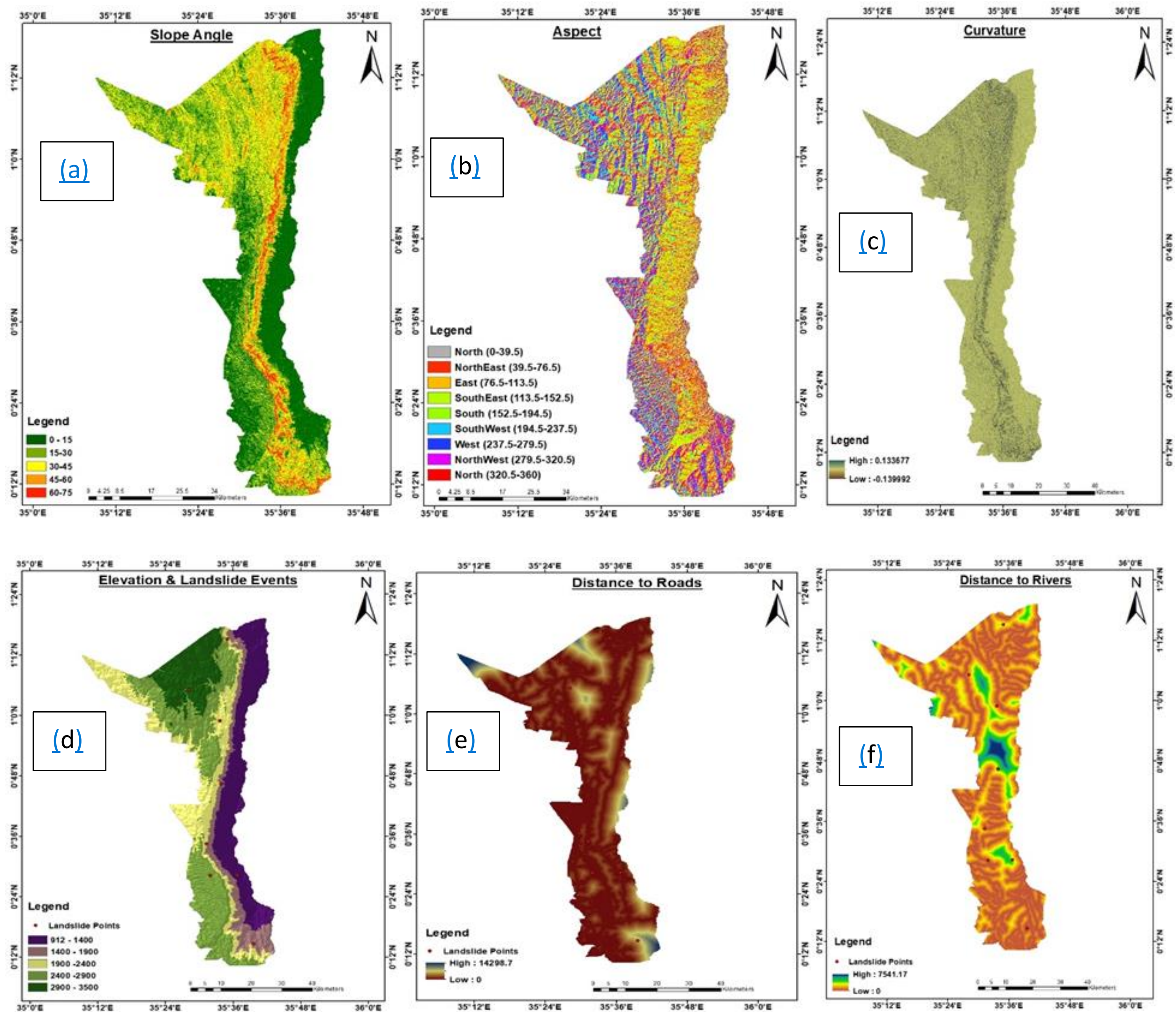
Table 1: Data sources and their resolutions

Data	Sources	Resolution
SRTM DEM (Elevation, slope, aspect, curvature, TWI)	USGS	30m
Landslides Inventory Data	Global Precipitation Measurement Mission	
Landsat 8 OLI	USGS	30m
Roads, Rivers & Streams	DIVA GIS, OSM	30m
Lithology	Water Resource Authority	30m



(a)
Slope Angle
Legend
0 - 15
15-30
30-45
45-60
60-75
(b)
Aspect
Legend
North (0-0.7)
North East (0.7-1.2)
East (1.2-1.9)
South East (1.9-2.7)
South (2.7-3.5)
South West (3.5-4.2)
West (4.2-5)
North West (5-5.7)
North (5.7-6.2)
(c)
Curvature
Legend
High : 0.165654
Low : -0.0896854
(d)
Elevation & Landslide Events
Legend
Landslide Points
698 - 1,101
1,101 - 1,470
1,470 - 1,885
1,885 - 2,492
2,492- 3,516
(e)
Distance to Roads
Legend
Landslide Points
High : 29341.5
Low : 0
(f)
Distance to Rivers
Legend
Landslide Points
High : 8903.8
Low : 0
(g)
NDVI
Legend
High : 0.530281
Low : -0.142041
(h)
Topographic Wetness Index
Legend
<5
5-10
10-15
15-20
>20
(i)
Lithology
Legend
PD
PV
MV2
Q2C
Q2

Figure 3: Landslide conditioning factors for source region a) Slope b)Aspect c)Curvature d) Elevation & landslide events e) Distance to Roads f) Distance to Rivers g)NDVI h)Topographic Wetness Index i)Lithology



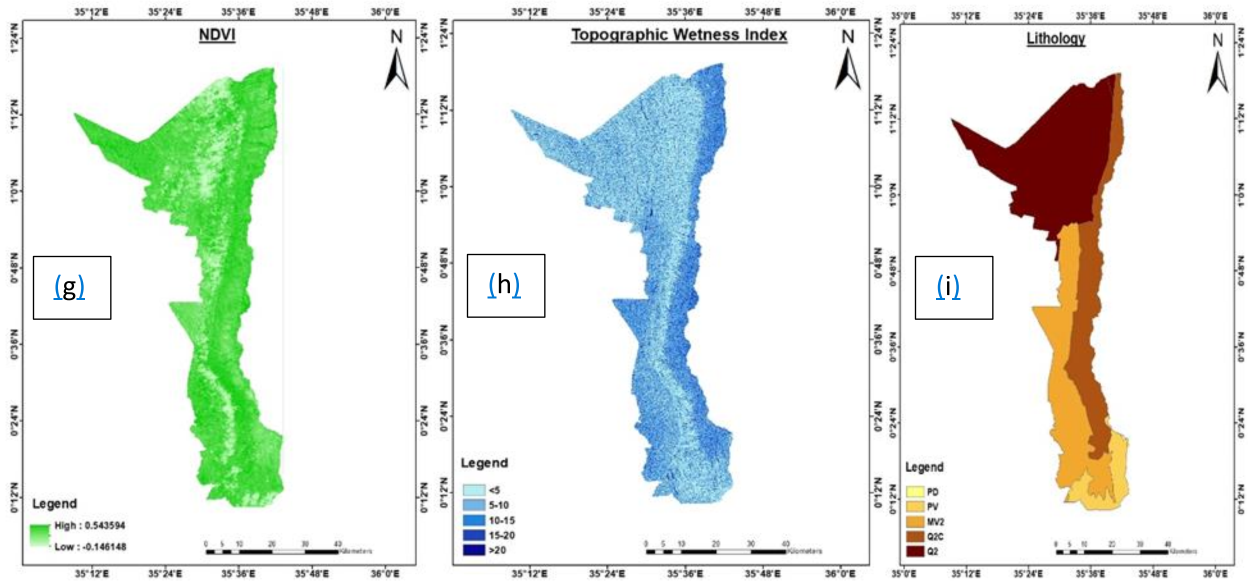


Figure 4: Landslide conditioning factors for target region a) Slope b)Aspect c)Curvature d) Elevation & landslide events e) Distance to Roads f) Distance to Rivers g)NDVI h)Topographic Wetness Index i)Lithology

2.3.4 Multicollinearity Analysis

Multicollinearity is a statistical issue that arises when two or more explanatory variables in a model are highly correlated (Chan et al., 2022). This strong interrelationship among variables can negatively affect the accuracy and reliability of statistical and machine learning models by making it difficult to isolate the individual impact of each predictor. In landslide susceptibility modeling, where several predisposing factors are often used, it is essential to check for multicollinearity before proceeding with model development. Failure to address multicollinearity can result in unstable estimates, inflated standard errors, and misleading interpretations.

To diagnose multicollinearity, several techniques are available. Common approaches include Pearson's correlation analysis, the Variance Inflation Factor (VIF), tolerance analysis, the Farrar–Glauber test, and the condition number test. Among these, VIF and tolerance are particularly popular in environmental and geospatial modeling due to their straightforward interpretation and effectiveness (Lee et al., 2020). It is calculated using the formula;

$$VIF = \frac{1}{Tolerance} = \frac{1}{1-R^2} \quad \text{.....................(3)}$$

Where, VIF is the Variance Inflation Factor and, R^2 is the coefficient of determination obtained by regressing one variable against all the others. A higher VIF indicates stronger multicollinearity, while tolerance, which is its reciprocal, decreases as multicollinearity increases.

2.3.5 Support Vector Machine (SVM) Model Training

The Support Vector Machine (SVM) algorithm was selected for landslide susceptibility modeling due to its ability to manage high-dimensional data and small sample sizes effectively. The dataset, composed of landslide conditioning factors and corresponding labels, was cleaned, normalized, and randomly split into training (70%) and testing (30%) sets. An SVM with a Radial Basis Function (RBF) kernel was used to capture non-linear relationships by mapping features into a higher-dimensional space, allowing the model to optimize class separation (Kamran et al., 2021).

2.3.6 Hyperparameter Optimization

To improve model generalization, key hyperparameters regularization parameter (C) and kernel coefficient (γ) were fine-tuned using grid search. The C parameter controls the balance between margin maximization and classification error, while γ determines the influence of individual training points. Grid search, combined with cross-validation, was employed to systematically evaluate different C and γ values, selecting the combination that maximized classification accuracy (Abbas et al., 2023).

2.3.7 Accuracy Assessment

Model performance was assessed using the reserved test dataset through both threshold-based and continuous metrics. These included accuracy, precision, recall, F1-score, Cohen’s kappa, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Precision and recall measured the model's correctness and completeness in detecting landslide events, while the F1-score provided a harmonic mean of both. Accuracy and kappa evaluated overall and chance-adjusted agreement, whereas RMSE and MAE quantified prediction error magnitudes.

$$Sensitivity = \frac{TP}{TP+FN} \quad \text{......................(4)}$$

$$Specificity = \frac{TN}{TN+FP} \quad \text{......................(5)}$$

$$Precision = \frac{TP}{TP+FP} \quad \text{.........................(6)}$$

$$Recall = \frac{TP}{TP+FN} \quad \text{...........................(7)}$$

$$F - score = \frac{2 * Precision * Recall}{Precision+Recall} \quad \text{.............(8)}$$

$$Accuracy = \frac{TP+TN}{TP+FN+TN+FP} \ldots\ldots\ldots\ldots\ldots(9)$$

Where TN = , FP = , FN

The summary of this study is depicted in figure 5.

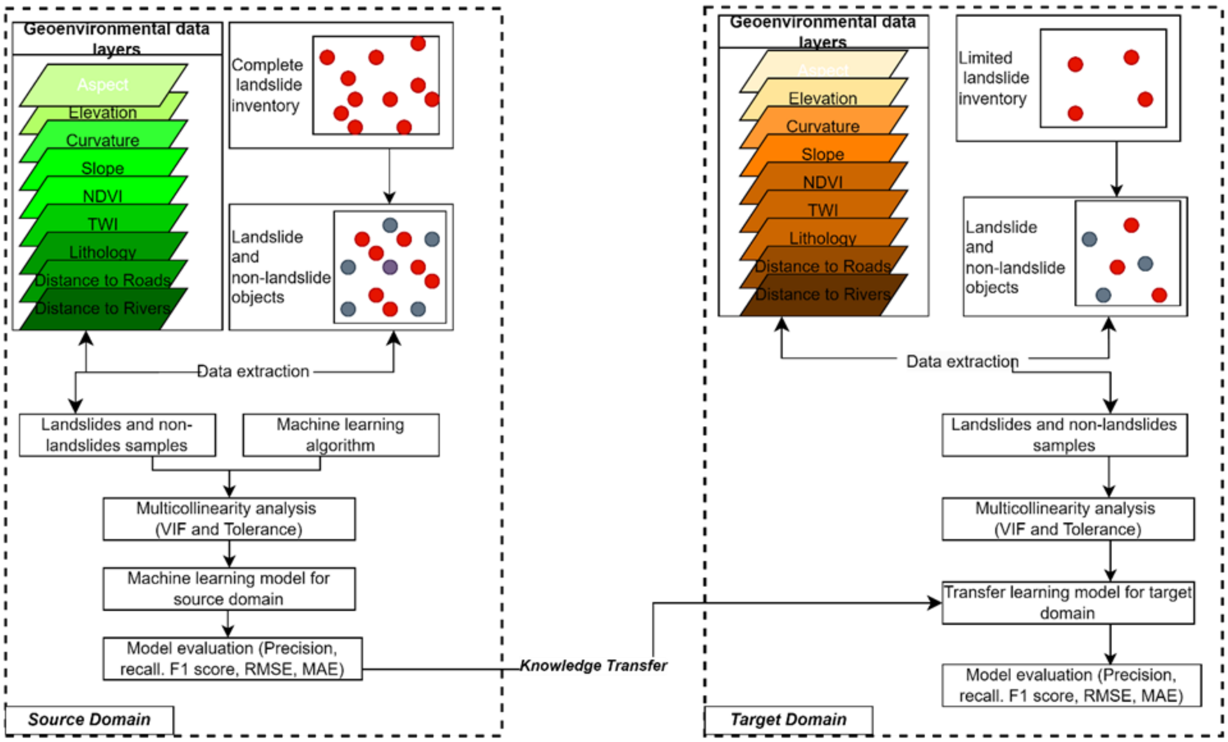


Figure 5: Summary of the study methodology workflow

3. Results

3.1 Multicollinearity assessment

The multicollinearity assessment for West Pokot and Elgeyo Marakwet showed that all predictor variables had Variance Inflation Factor (VIF) values below 10, indicating that multicollinearity is within acceptable limits. Although some variables recorded relatively higher VIF values, none exceeded the threshold where corrective action (such as variable elimination) would be necessary.

Table 2: The multicollinearity assessment results for West Pokot County

WEST POKOT MULTICOLLINEARITY

Variable	Tolerance	VIF
Slope	0.2495	4.0073
Aspect	0.2012	4.9681
Curvature	0.6394	1.5639
Elevation	0.3986	2.5089
NDVI	0.5600	1.7856
TWI	0.4494	2.2251
Distance to Roads	0.6581	1.5193
Distance to Rivers	0.5829	1.7155
Lithology	0.9657	1.0355

Table 3: The multicollinearity assessment results for Elgeyo Marakwet County

ELGEYO MARAKWET MULTICOLLINEARITY		
Variable	**Tolerance**	**VIF**
Slope	0.1970	5.0754
Aspect	0.1498	6.6745
Curvature	0.1496	6.6854
Elevation	0.4534	2.2052
NDVI	0.6517	1.5344
TWI	0.4163	2.4018
Distance to Roads	0.1551	6.4461
Distance to Rivers	0.6959	1.4370
Lithology	0.2758	3.6261

3.2 Hyperparameter Optimization

To optimize the performance of the SVM model, a grid search was conducted over a range of values for the regularization parameter (C) and the kernel coefficient (gamma). The goal was to identify the combination that minimizes the classification error.From the tuning results, the lowest error rate was observed when C = 10 and gamma = 0.1(Figure 6). This combination strikes a balance between underfitting and overfitting.

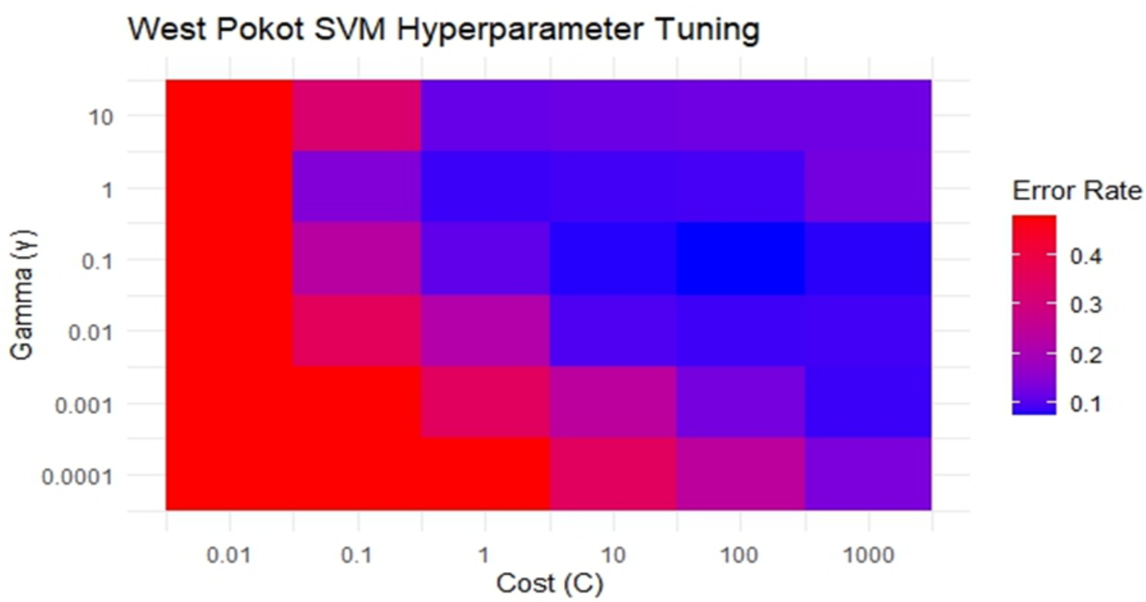


Figure 6: Plot showing SVM hyperparameter tuning for West Pokot County

3.3 Variable Importance in SVM Model

The variable importance plot reveals that Elevation is by far the most influential factor in predicting landslide susceptibility in West Pokot (Figure 7). It accounts for over 50% of the model's relative importance, indicating that elevation variations play a significant role in landslide occurrence across the region. Whereas the variable importance results for Elgeyo Marakwet (Figure 8) reveal that Distance to Rivers is the most critical factor in predicting landslide susceptibility, with the highest relative importance (approximately 60%). This strong influence suggests that areas closer to rivers are significantly more prone to landslides, likely due to erosion, prolonged soil saturation, and the undercutting of slopes by flowing water.

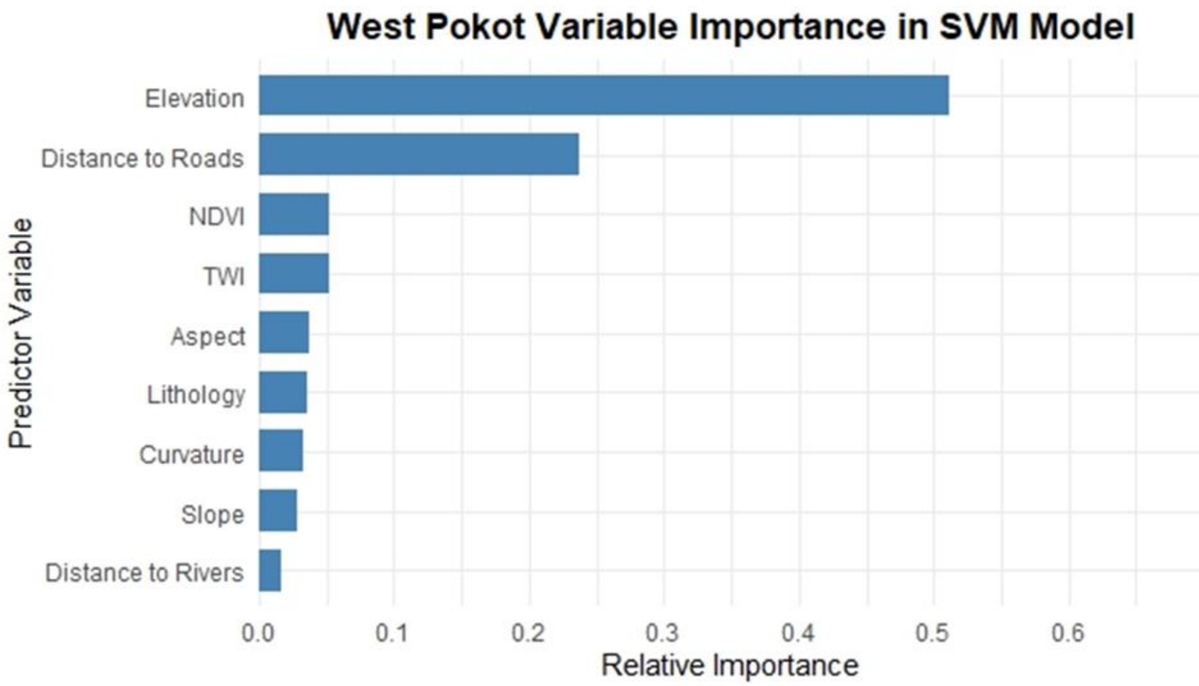


Figure 7: West Pokot Variable Importance in SVM Model

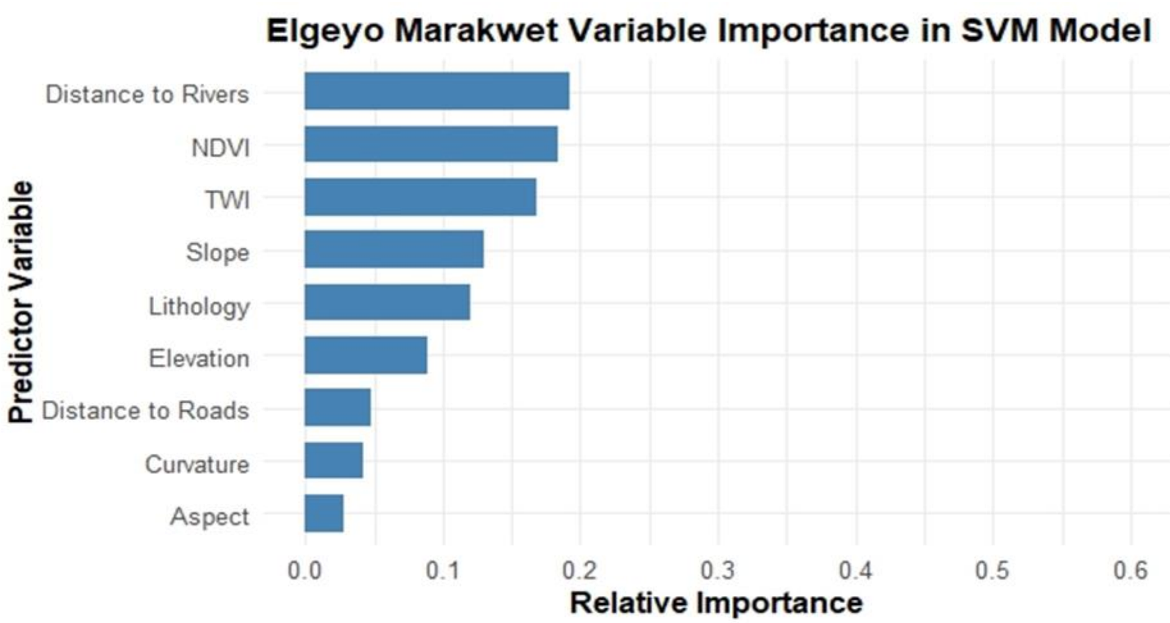


Figure 8: Elgeyo Marakwet Variable Importance in SVM Model

3.4 Landslide Susceptibility and affected areas Maps

The susceptibility map for West Pokot (Figure 9a) reveals spatial variability in landslide risk across the county. Areas classified as *Very High* and *High* susceptibility are prominently concentrated in the central and southeastern regions, particularly around locations such as Chepareria, Tapach, and Kodich. These zones correspond to regions with steep slopes, dense drainage networks, and significant land cover changes key factors contributing to landslide occurrence.On the other hand, the *Moderate* and *Low* susceptibility zones dominate the northern and western parts of the county, which generally exhibit gentler slopes and more stable terrain.

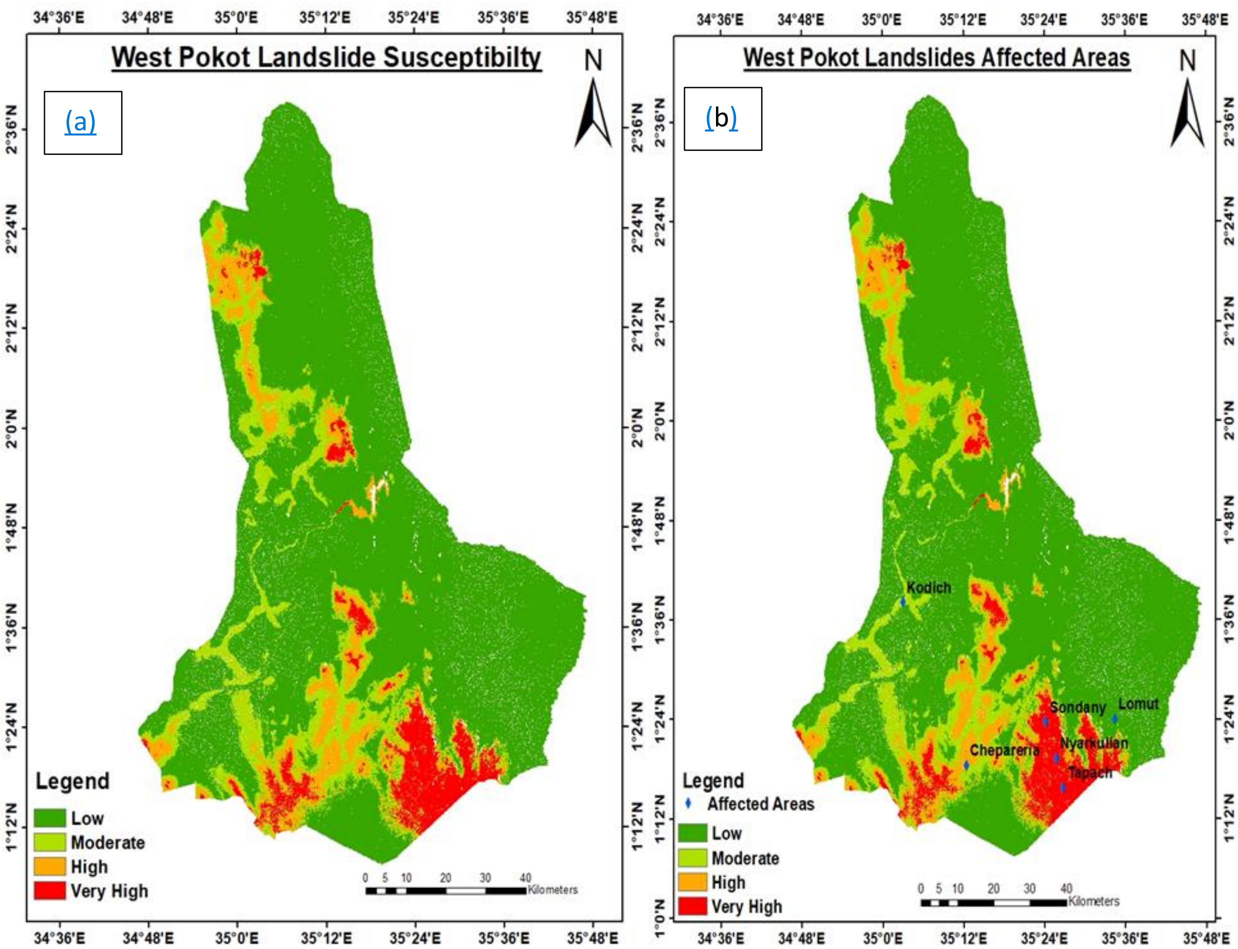


Figure 9: a) Landslide susceptibility map for West Pokot County b) Landslide affected areas map for West Pokot County

The susceptibility map for Elgeyo Marakwet (Figure 10a) reveals distinct spatial patterns in landslide risk across the county. Areas classified as Very High and High susceptibility are predominantly concentrated in the central and northern parts of the county, particularly in wards such as Embobut, Arror, Kaben, and Chesoi. These zones align with regions characterized by steep terrain, proximity to river systems, and unstable geological formations factors that significantly elevate landslide vulnerability.

In contrast, Moderate and Low susceptibility zones are more prevalent in the southern and eastern parts of the county, including areas like Kaptarakwa and Emsoo Ward. These regions generally feature gentler slopes, more stable lithology, and lower hydrological activity, resulting in reduced landslide risk.

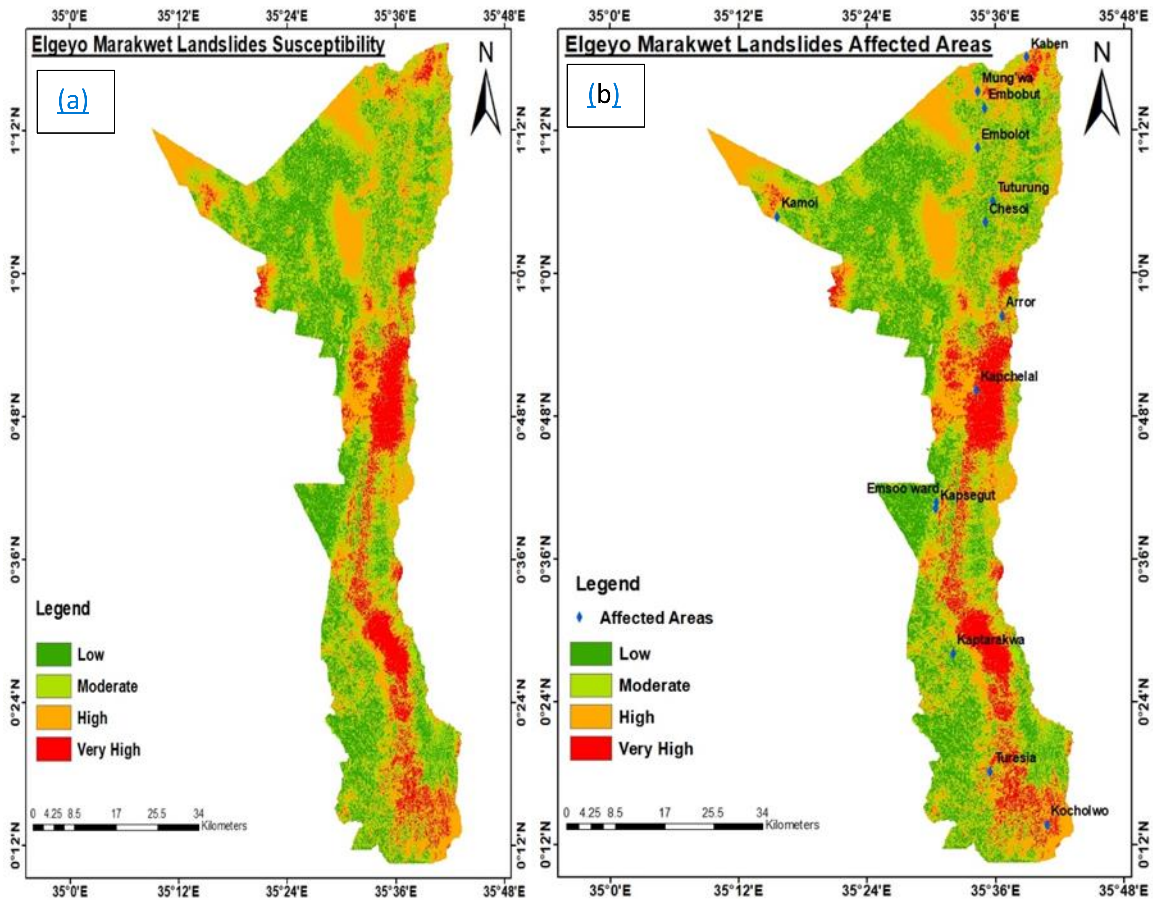


Figure 10: a) The susceptibility map for Elgeyo Marakwet County b) Landslide affected areas map for Elgeyo Marakwet County

3.5 Accuracy Assessment

The model’s performance in West Pokot, the source region, was robust across all evaluation metrics. A high precision score of 0.9304 indicates that most of the predicted landslide-prone areas were correctly identified. The recall score of 0.8917 shows the model's ability to capture actual landslide events, while the F1 score of 0.9106 reflects a strong balance between precision and recall. Overall, the model achieved an accuracy of 91.53% and a Cohen’s kappa coefficient of 0.8302, suggesting strong agreement beyond random chance. Additionally, the low RMSE (0.291) and MAE (0.0847) values confirm minimal prediction error, further validating the model’s reliability in the source domain.

In Elgeyo Marakwet, the target region, the transferred model maintained a high precision of 0.9536, showing that most of the predicted landslide zones were accurate. However, recall decreased to 0.6667, suggesting that the model missed a significant number of actual landslides. Consequently, the F1 score dropped to 0.80, and accuracy fell to 83.33%. The kappa coefficient (0.6667) indicates moderate agreement, while increased RMSE (0.4679) and MAE (0.4638) values reflect higher error margins. This performance gap between regions demonstrates the promise of transfer learning but also highlights the limitations of domain shift due to environmental variability.

Table 3: Accuracy assessment results

	Precision	Recall	F1 Score	Accuracy	Kappa	RMSE	MAE
West Pokot	0.9304	0.8917	0.9106	0.9153	0.8302	0.291	0.0847
Elgeyo Marakwet	0.9536	0.6667	0.80	0.8333	0.6667	0.4679	0.4638

4. Discussion

The results of this study validate the use of transfer learning (TL) for landslide susceptibility modeling in data-constrained environments, specifically highlighting its effectiveness in cross-regional applications between West Pokot (source) and Elgeyo Marakwet (target). The SVM model trained on the rich landslide inventory of West Pokot performed with reasonable generalization in Elgeyo Marakwet, achieving an accuracy of 83.33% and an F1-score of 80.0%. These outcomes are comparable to prior studies that reported similar transferability using TL frameworks in mountainous or data-scarce terrains (Zhang et al., 2024; Singh et al., 2024).

In West Pokot, the most influential landslide conditioning factors (LCFs) were Elevation and Distance to Roads. This can be attributed to the county's rugged topography and extensive human-induced modifications to slopes, especially along poorly engineered rural roads. Such disturbances often lead to water concentration and increased pore pressure during rainfall events, thus reducing slope stability. These findings align with Al-Najjar et al. (2019), who emphasized the role of anthropogenic infrastructure in increasing landslide susceptibility in tropical highlands.

In contrast, in Elgeyo Marakwet, Distance to Rivers was the dominant predictor. The county’s escarpment terrain, coupled with significant fluvial activity from the Kerio River and its tributaries, contributes to toe erosion, saturation of slope materials, and eventual mass movements. This regional divergence in influential variables supports the notion that TL models must account for localized geomorphological processes even when generalized across similar physiographic zones. Simiyu et al. (2022) also observed that in Elgeyo Marakwet, river-induced erosion plays a critical role in triggering landslides, especially in Marakwet East and Keiyo South.

Moreover, the variable importance rankings obtained through the SVM model further reveal that while the same set of predictors was used across both regions, their individual influence on slope failure varies significantly. This underscores the importance of interpreting transferred models with domain knowledge. For instance, while NDVI and TWI were moderately ranked in both regions, their hydrological influence may manifest differently due to microclimatic and soil structure variations (Kibet, 2021).

Another notable finding is the performance gap between source and target regions. The SVM model achieved superior metrics in West Pokot (F1-score: 91.06%, Accuracy: 91.53%) compared to Elgeyo Marakwet. This discrepancy reflects the challenge of domain shift—a phenomenon where the statistical distribution of variables differs between domains. Such a shift can impair the ability of a transferred model to perfectly replicate performance across regions. Zhao et al. (2024) argue that this limitation can be mitigated through fine-tuning, which involves updating the model with limited labeled data from the target region to recalibrate its decision boundaries.

The acceptable multicollinearity levels in both counties (all VIF values below 10) enhance the credibility of the model output. As emphasized by Chan et al. (2022), maintaining low inter-correlation among predictors ensures that each LCF contributes unique and meaningful information to the model.

From a practical standpoint, this research contributes a transferable and cost-effective framework for generating landslide susceptibility maps in under-resourced counties. The ability to apply a model trained in West Pokot to Elgeyo Marakwet without the need for extensive local inventories addresses a critical challenge in geohazard modeling in sub-Saharan Africa. As Donnini et al. (2020) assert, integrating ML-based susceptibility maps into disaster risk management systems can improve local preparedness, zoning regulations, and infrastructure planning.

Finally, while this study demonstrates the strength of TL and SVM in landslide modeling, it also highlights areas for improvement. Future studies should consider incorporating fine-tuning techniques using a small sample of local data in the target region, and experimenting with ensemble learning approaches to combine predictions from multiple algorithms. The addition of time-dependent variables such as rainfall and real-time monitoring through IoT could further enhance the dynamic responsiveness of the model.

5. Conclusion

This study successfully applied a cross-regional transfer learning framework to develop a landslide susceptibility model using SVM, with West Pokot as the source and Elgeyo Marakwet as the target. The research demonstrated that TL is a powerful technique for extending the utility of machine learning models across regions with similar environmental and geological contexts.

The model achieved high accuracy and reliability in both regions, with slight reductions in performance in the target area due to localized terrain dynamics. Nonetheless, the success of the transferred model indicates that TL can serve as a practical solution in scenarios where historical landslide data are limited or nonexistent.

In the future, incorporating fine-tuning techniques, where limited local data are used to slightly adjust the source model, could further enhance accuracy. Additionally, integrating time-dependent variables such as rainfall and land-use change, alongside real-time IoT-based monitoring systems, could improve the dynamic responsiveness of susceptibility models.

As machine learning continues to evolve, combining TL with emerging deep learning architectures and ensemble models could provide even greater accuracy and flexibility in diverse environmental contexts.

Funding

The present study received no funding.

Data availability

The datasets used to generate the findings of this study are available from the corresponding author upon a reasonable request.

Declarations

All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors.

Ethics approval

Not applicable.

Consent to participate

Not applicable.

Consent for publication

The authors warrant that the work has not been published before in any form and is not under consideration by another publisher, that the persons listed above are in the proper order, that no author entitled to credit has been omitted, and generally that the authors have the right to make the grants made to the publisher complete and unencumbered. The authors also warrant that the work does not libel anyone, infringe anyone's copyright, or otherwise violate anyone's statutory or common law rights.

Competing interests

The authors declare no competing interests.

References

Abbas, F., Zhang, F., Ismail, M., Khan, G., Iqbal, J., Alrefaei, A. F., & Albeshr, M. F. (2023). Optimizing machine learning algorithms for landslide susceptibility mapping along the Karakoram Highway, Gilgit Baltistan, Pakistan: A comparative study of baseline, bayesian, and metaheuristic hyperparameter optimization techniques. *Sensors*, *23*(15), 6843. https://www.mdpi.com/1424-8220/23/15/6843

Agboola, G., Beni, L. H., Elbayoumi, T., & Thompson, G. (2024). Optimizing landslide susceptibility mapping using machine learning and geospatial techniques. *Ecological Informatics*, *81*, 102583. https://www.sciencedirect.com/science/article/pii/S1574954124001250

Al-Najjar, H. A., Kalantar, B., Pradhan, B., & Saeidi, V. (2019, October). Conditioning factor determination for mapping and prediction of landslide susceptibility using machine learning algorithms. In *Earth resources and environmental remote sensing/GIS applications X* (Vol. 11156, pp. 97-107). SPIE. https://www.spiedigitallibrary.org/conference-proceedings-of-spie/11156/111560K/Conditioning-factor-determination-for-mapping-and-prediction-of-landslide-susceptibility/10.1117/12.2532687.short

Asadi, A., Baise, L. G., Chatterjee, S., Koch, M., & Moaveni, B. (2024). Regional landslide mapping model developed by a deep transfer learning framework using post-event optical imagery. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, *18*(1), 186-210. https://www.tandfonline.com/doi/abs/10.1080/17499518.2024.2316265

Chan, J. Y. L., Leow, S. M. H., Bea, K. T., Cheng, W. K., Phoong, S. W., Hong, Z. W., & Chen, Y. L. (2022). Mitigating the multicollinearity problem and its machine learning approach: a review. *Mathematics*, *10*(8), 1283. https://www.mdpi.com/2227-7390/10/8/1283

Donnini, M., Modica, M., Salvati, P., Marchesini, I., Rossi, M., Guzzetti, F., & Zoboli, R. (2020). Economic landslide susceptibility under a socio-economic perspective: an application to Umbria Region (Central Italy). *Review of Regional Research*, *40*, 159-188. https://link.springer.com/article/10.1007/s10037-020-00143-6

Guo, Z., Tian, B., Zhu, Y., He, J., & Zhang, T. (2024). How do the landslide and non-landslide sampling strategies impact landslide susceptibility assessment?—A catchment-scale case study from China. *Journal of Rock Mechanics and Geotechnical Engineering*, *16*(3), 877-894. https://www.sciencedirect.com/science/article/pii/S1674775523003608

Kamran, K. V., Feizizadeh, B., Khorrami, B., & Ebadi, Y. (2021). A comparative approach of support vector machine kernel functions for GIS-based landslide susceptibility mapping. *Applied Geomatics*, *13*(4), 837-851. https://link.springer.com/article/10.1007/s12518-021-00393-0

Kibet, S. J. (2021). *The role of forest cover in landslide risk reduction in Marakwet East Escarpment, Elgeyo-Marakwet County, Kenya* [Master's thesis, University of Nairobi]. University of Nairobi Repository. http://erepository.uonbi.ac.ke/handle/11295/160982

Laigle, D., & Bardou, E. (2022). Mass-movement types and processes: Flow-like mass movements, debris flows and earth flows. *Treatise on Geomorphology 2nd Edition*, 61-84. https://hal.science/hal-04430362/

Lee, D. H., Kim, Y. T., & Lee, S. R. (2020). Shallow landslide susceptibility models based on artificial neural networks considering the factor selection method and various non-linear activation functions. *Remote Sensing*, *12*(7), 1194. https://www.mdpi.com/2072-4292/12/7/1194

Lin, Z., Chen, Q., Lu, W., Ji, Y., Liang, W., & Sun, X. (2023). Landslide Susceptibility Mapping Based on Information-GRUResNet Model in the Changzhou Town, China. *Forests*, *14*(3), 499. https://www.mdpi.com/1999-4907/14/3/499

Meena, S. R., Puliero, S., Bhuyan, K., Floris, M., & Catani, F. (2022). Assessing the importance of conditioning factor selection in landslide susceptibility for the province of Belluno (region of Veneto, northeastern Italy). *Natural hazards and earth system sciences*, *22*(4), 1395-1417. https://nhess.copernicus.org/articles/22/1395/2022/nhess-22-1395-2022.html

Namwanje, R. (2021). *A comparative analysis on GIS-based methods and machine learning algorithm in landslide susceptibility modeling: A case study of Bududa, Uganda* [Master’s thesis, Pan African University Institute of Water and Energy Sciences]. http://repository.pauwes-cop.net/handle/1/492

Pánek, T. (2020). Landslides and related sediments. *Encyclopedia of Geology,*, *2*, 708-728. https://www.researchgate.net/profile/Tomas-Panek-3/publication/341535831_Landslides_and_Related_Sediments/links/5ee4a65492851ce9e7e05921/Landslides-and-Related-Sediments.pdf

Parkash, S. (2023). Lessons learned from landslides of socio-economic and environmental significance in india. In *Progress in Landslide Research and Technology, Volume 1 Issue 2, 2022* (pp. 309-315). Cham: Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-031-18471-0_23

Shahabi, H., Rahimzad, M., Tavakkoli Piralilou, S., Ghorbanzadeh, O., Homayouni, S., Blaschke, T., ... & Ghamisi, P. (2021). Unsupervised deep learning for landslide detection from multispectral sentinel-2 imagery. *Remote Sensing*, *13*(22), 4698. https://www.mdpi.com/2072-4292/13/22/4698

Shrestha, M., Sharma, S., & Pradhan Shrestha, R. (2025). Landslides in the Himalayas: A Comprehensive Review of Hazards, Impacts, and Adaptive Strategies. *Rural and Regional Development*, *3*(1), 10002. https://www.sciepublish.com/article/pii/408

Sim, K. B., Lee, M. L., & Wong, S. Y. (2022). A review of landslide acceptable risk and tolerable risk. *Geoenvironmental Disasters*, *9*(1), 3. https://link.springer.com/article/10.1186/s40677-022-00205-6

Simiyu, J. N., Wambua, B. N., & Njoka, J. T. (2022). Hydrological drivers of slope failure in Elgeyo Marakwet: An integrated GIS approach. *Natural Hazards, 111*(2), 1021–1045. https://doi.org/10.1007/s11069-021-05005-9

Singh, A., Dhiman, N., & Shukla, D. P. (2024, July). Transfer Learning in Landslide Susceptibility Mapping: Bridging Data-Rich and Data-Scarce Regions in the Northwestern Himalayas. In *IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium* (pp. 3253-3256). IEEE. https://ieeexplore.ieee.org/abstract/document/10641843/

Singh, A., Dhiman, N., Niraj, K. C., & Shukla, D. P. (2024). Ensembled transfer learning approach for error reduction in landslide susceptibility mapping of the data scare region. *Scientific Reports*, *14*(1), 29060. https://www.nature.com/articles/s41598-024-76541-4

Wang, H., Wang, L., & Zhang, L. (2023). Transfer learning improves landslide susceptibility assessment. *Gondwana Research*, *123*, 238-254. https://www.sciencedirect.com/science/article/pii/S1342937X22002131

Wang, X., Wang, D., Li, X., Zhang, M., Cheng, S., Li, S., ... & Huang, X. (2024). The prediction of cross-regional landslide susceptibility based on pixel transfer learning. *Remote Sensing*, *16*(2), 347. https://www.mdpi.com/2072-4292/16/2/347

Wang, Z., & Brenning, A. (2021). Active-learning approaches for landslide mapping using support vector machines. Remote Sensing, 13(13), 2588. https://www.mdpi.com/2072-4292/13/13/2588

Wang, Z., Goetz, J., & Brenning, A. (2022). Transfer learning for landslide susceptibility modelling using domain adaptation and case-based reasoning. *Geoscientific Model Development Discussions*, *2022*, 1-30. https://gmd.copernicus.org/articles/15/8765/2022/

Zhang, Y., Wang, L., & Li, H. (2024). Enhanced landslide susceptibility mapping in data-scarce regions via unsupervised few-shot learning. *Landslides, 21*(3), 789–803. https://doi.org/10.1016/j.landslides.2024.03.006

Zhao, Z., Wang, W., Chen, J., Yao, J., Liao, Y., & Liu, J. (2024). Heterogeneous transfer learning considering feature representation and environmental consistency for landslide spatial prediction. *GIScience & Remote Sensing*, *61*(1), 2349343. https://www.tandfonline.com/doi/abs/10.1080/15481603.2024.2349343